

Liancheng Gong (Krystal)

CONTACT INFORMATION	College Park MD 20742	lianchenggong0224@gmail.com gonglc@umd.edu
EDUCATION	University of Maryland College Park, MD Doctor of Philosophy in Information Studies August 2025 - present Advised by Assistant Professor Julia Mendelsohn and Assistant Professor Wei Ai	
	University of Pennsylvania Philadelphia, PA Master of Science in Engineering in Data Science, GPA: 3.90/4.00 August 2022 - May 2024 • Math: <i>Big Data Analytics, Statistics for Data Science</i> • Computer Science: <i>Database and Information Systems, Intro to Software Develop</i> • Artificial Intelligence: <i>Principles of Deep Learning, Artificial Intelligence, Ethical Algorithm Design</i>	
	New York University New York City, NY Abu Dhabi, UAE Shanghai, China Bachelor of Science in Data Science, GPA: 3.81/4.00 August 2018 - May 2022 • Math: <i>Probability and Statistics, Ordinary Diff Equations, Introduction to Math Modeling</i> • Core Data Science: <i>Data Structures, Machine Learning, Information Visualization, NLP, Algorithms</i> • Human-Centered: <i>Decision Making Under Uncertainty, Human-Centered Data Science</i>	
HONORS AND AWARDS	Dean’s Fellowship, University of Maryland	FY26
	Magna Cum Laude, New York University	2022
	Dean’s List for Academic Year, New York University	2019, 2020, 2021, 2022
PUBLICATIONS	• Liancheng Gong, Wang Zhu, Jesse Thomason, and Li Zhang. <i>Zero-Shot Iterative Formalization and Planning in Partially Observable Environments</i>. arXiv preprint <i>arXiv:2505.13126</i>, 2025. [Link] [under review] • Liancheng Gong, Daniel J. Hopkins, Samuel Wolken. <i>The Limits of De-politicizing—and Also of Annotation: A Case Study in Russian Newspaper Posts on Social Media 2016–2024</i> [under review]	
REVIEW	• AAI • IC2S2, ACM TIST	2025 2026
FUNDING	• 2026 CRA-WP Grad Cohort Workshop Funding source: CRA Scholarship	
RESEARCH EXPERIENCE	Computer Science department, Drexel University Philadelphia, PA <i>Researcher</i> working with Assistant Professor Harry Zhang September 2024 - July 2025 • Structured Large Language Models (LLM) outputs into Planning Domain Definition Language (PDDL) domain and problem files to generate verifiable plans • Iteratively refined outputs using environmental feedback to improve generation accuracy • Categorized recurring errors across trials, experimenting with solutions including memory integration and partial file editing to enhance model robustness	
	Political Science department, University of Pennsylvania Philadelphia, PA <i>Research Coordinator (full-time)</i> working with Professor Daniel Hopkins August 2024 - July 2025 • Analyzed polls and exit polls for FiveThirtyEight’s 2024 presidential election coverage	

- Investigated Democratic and Republican advertisement campaigns across Meta, Instagram, and Google; examined topic trends in Russian media outlets, such as anti-government narratives
- Scraped and processed 2M+ VKontakte posts from 19 media outlets via **HPC**, created feature vectors using **TF-IDF and BERT** embeddings, developed a **logistic regression model** to identify political posts, increasing the sample's political content from 10% to 50%
- Annotated Russian blurbs via **OpenAI API**, including prompt engineering for annotation and translation, achieving 90% alignment between GPT and human annotations on political features
- Classified symbolic rhetoric by generating embeddings and fine-tuning on **BERTweet** to detect topics like “us vs them” and abstract themes

Computational Social Science Lab, University of Pennsylvania

Philadelphia, PA

Data Scientist working with **Yuxuan Zhang** and **Regina (Jeanne) Ruane** February 2024 - May 2024

- Designed a master list to annotate and analyze bias in media discussions on topics like Biden's age and NATO, delivering findings to the **dashboard**
- Automated **extract, transform, and load (ETL)** processes for streaming Nielsen data, improving data consistency and minimizing manual handling
- Managed an **AWS-hosted database**, ensuring data accuracy and schema consistency; standardized network names across large datasets for enhanced data cohesion and analysis

Graduate Research Assistant

February 2023 - December 2023

Atlas working with **Mark Whiting**

- Segmented and mined information from PDFs, leveraging **OpenAI's question-answering API** and context window techniques, e.g. Dynamic Memory Networks, Parallel Context Windows
- Pioneered **LLMs** approach by refining prompts and GPT models, achieving superior comprehension and context alignment
- Sourced a robust dataset of 300+ papers and crafted 300+ precise questions, covering all critical analysis units and elevating the accuracy and depth of the literature analysis

Human Mobility Data for Epidemiology working with **Jorge Barreras Cortes**

- Developed **Susceptible-Exposed-Infected-Recovered models** on contact networks utilizing **Poisson Distribution** and **Tensor** to predict virus spread, guiding intervention strategies
- Managed **ETL** of 100GB+ databases via **AWS RDS and EMR**; conducted cross-analysis on POI distribution and mobility indicators, ensuring model's applicability and accuracy
- Optimizing the **DBSCAN** clustering algorithm on population data, providing insights of area-specific visit patterns

PROJECTS

Machine Q&A on Chinese Judicial Reading Comprehension

[Code], [PDF]

Developed a Span-Extraction Machine Reading Comprehension (MRC) system

Colorize images with Deep Learning

[Code], [PDF]

Compared ResNet, GAN, VAE to colorize greyscale images

Customer Lifecycle, NYU X HSBC

[Code], [PDF]

Segmented customers, predicted inactivity, and provided targeted recommendations for HSBC

A Model to Predict the Maximum Car Speed in the Rain

[Code], [PDF]

Quantified the visibility of the driver and calculate maximum speed under safe speed consideration

TEACHING EXPERIENCE

Fife Academy Coding Club, University of Pennsylvania

Philadelphia, PA

Coding Instructor

October 2022 - May 2023

- Developed and led a modular coding curriculum for 100+ K-12 students, with a 100% positive feedback rate, delivering weekly lectures, hands-on labs, and individual feedback

WORKING EXPERIENCE

Tencent Technology (Shenzhen) Co., Ltd

Shanghai, China

Data Scientist, Platform and Content Group

May 2021 - July 2021

- Enhanced user segmentation and ad CTR by 5% through analysis of 100M+ user data (Python, MySQL); developed ARIMA (0.073 RMSE) for activity forecasting and used NLP on search logs, increasing retention by 4% via A/B testing

Shanghai Urban Real Estate Appraisal Co., Ltd

Shanghai, China

Data Analyst, Data R&D department

December 2020 - April 2021

- Developed a predictive model (Regression, Random Forest, XGBoost) with 80% R^2 for rental pricing, supported by literature review, survey (300+ responses), and a 2000+ entry database from field visits and web scraping

COMPUTER SKILLS

- Programming Languages: Python, R, SQL, Java
- Frameworks and Libraries: NumPy, Pandas, SciPy, Scikit-Learn, PySpark, Seaborn, Matplotlib, PyTorch, TensorFlow
- Tools and platforms: AWS (RDS/Sagemaker/S3/EMR), MySQL, MongoDB, Tableau, PowerBI, Git, Linux, Stata, Matlab, ETL